Discrete Math

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— DM ~ Physics.

I Intro   II Propositional Logic   III ~~Logistics~~

"Discrete mathematical objects".
- not continuous (real numbers, vectors, calculus).
- integers, cards, dice, networks.

Previous Courses (51, 54, ...)

Emphasis on learning and using existing math.

- $\frac{d}{dx}(fg) = f \cdot \frac{d}{dx} g + \frac{d}{dx} f \cdot g.$
- $\int_0^x f'(t)\,dt = f(x)$
- $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}.$

This Course: discovering and creating math.

Reading & writing proofs.

A proof is a structured form of writing used to establish mathematical truths and communicate them.

cf:

physics	~	experiments.
statistics, social sciences	~	experiments, observations.
law	~	evidence, jury, trial
"absolute truth." math	~	proof

+ Don't need calculus.
+ More creativity.
+ More writing, less equations.

II Propositional Logic

"Mathematics is the science that draws necessary conclusions."
— B. Peirce.

i.e. Given the truth of some statements, what other statements **must** be true?

Defn: A *proposition* is a declarative sentence which is either true or false, but not both.

truth value

sentence	proposition	Truth value
pigs can fly.	Y	F
Wow!	N	—
$2+2=4$.	Y	T
$x+5=9$.	N	— (depends on x)
The number of people in this room is odd.	Y	?

Remark: Must be 100% precisely specified,
i.e. must agree on definitions.

Q) How are the truth values of propositions related?
Will use letters ($p, q, \ldots$) to denote propositions.

Logical Connectives ("algebra" of propositions) (Defns.)

① **Negation** (NOT) : $\neg p$:= "not p."
$\neg p$ has the opposite truth value of p.

② **Conjunction** (AND) : $p \wedge q$:= "p and q."
$p \wedge q$ has value $\begin{cases} T & \text{if both } p, q \text{ are } T \\ F & \text{if at least one is } F \end{cases}$

ex: "pigs can fly and 2+2=4." is F.

③ **Disjunction** (OR) : $p \vee q$:= "p or q."
$p \vee q$ has truth value $\begin{cases} T & \text{if at least one is } T \\ F & \text{if both } p, q \text{ are } F. \end{cases}$

(Q) $p \oplus q = T$ iff one of p, q is true (XOR).

④ **Conditional**: $p \rightarrow q$:= "if p then q." "p is sufficient for q."
"p implies q."

$p \rightarrow q$ is $\begin{cases} T \text{ if } p \text{ is } T \text{ and } q \text{ is } T \\ T \text{ if } p \text{ is } F \text{ and } q \text{ is } T \text{ or } F. \\ F \text{ if } p \text{ is } T \text{ and } q \text{ is } F. \end{cases}$

p: "hypothesis" q: "conclusion"

Metaphor: $p \rightarrow q$ is a contract saying
if the hypothesis p is T
then the conclusion q is T.

$p \rightarrow q$ is T if the contract is satisfied.
F if " " violated.

ex: $p \to q$: "pigs can fly" $\longrightarrow$ "$2+2=4$." T

p, F $\qquad$ q, T

$q \to p$: "$2+2=4$" $\longrightarrow$ "pigs can fly." F

T $\qquad$ F

Remark:

- $p \to q$ is not the same as $q \to p$.
- $p \to q$ does <u>not</u> imply "causality".

  it is a relation between truth values.

Terminology: An expression obtained by combining propositional vars + connectives with parentheses is called a <u>compound proposition</u>.

ex: $((p \to q) \vee (q \to \neg r)) \wedge (r \to \neg q)$.

ex: $\neg q \vee p$.

Useful Device: truth table:

p	q	$\neg q$	$\neg q \vee p$	$q \to p$
T	T	F	T	T
T	F	T	T	T
F	T	F	F	F
F	F	T	T	T

Defn: Two compound propositions C_1, C_2 are equivalent, denoted $C_1 \equiv C_2$ if they have the same truth value for all truth values of the variables.

ex: $\neg q \vee p \equiv q \to p$

$\neg p \vee q \equiv p \to q$.

Remark: Like $(a+b)^2 = a^2+2ab+b^2$.

Terminology: A compound proposition

$C \equiv T$ is a **tautology**

$C \equiv F$ is a **contradiction**

Poll

ex: $(p \to q) \vee (q \to r)$. Is this a tautology, contradiction, or neither?

Method 1: (Truth table)

Mechanically fill in.

p	q	r	$p \to q$	$q \to r$	$(p\to q)\vee(q\to r)$
					T
					T
					T
					T
					T
					T
					T
					T

Tautology.

Method 2: There are 2 possibilities for the truth value of q: ← creativity.

Case 1: q is T.

$\therefore$ $p \to q$ is T. (defn of $\to$)

$\therefore$ $(p \to q) \vee (q \to r)$ is T. (defn $\vee$)

Defn of prop.

Case 2: q is F.

$\therefore$ $q \to r$ is T. (defn of $\to$)

$\therefore$ $(p \to q) \vee (q \to r)$ is T. (defn $\vee$)

$\therefore$ It is a tautology. $\square$

One more connective:

(5) Biconditional. $p \leftrightarrow q$:= "p if and only if q."

T if p and q have the same val.

F if " " different val.

<u>Remark:</u>

- $p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$.
- $(p \leftrightarrow q) \leftrightarrow ((p \to q) \wedge (q \to p)) \equiv T$.

Logic Puzzle: There is an island w/ Knights & Knaves.

1. Knights always tell the truth.
2. Knaves always lie.
3. Alice says "Bob and I are the same type."

(Q) Can you figure out what Bob is?

Model using propositional logic. (translate English to logic.)

Define: p : Alice is a Knight. $\neg p$: Alice is a knave.

q : Bob is a knight. $\neg q$: Bob is a knave.

Assumptions: 1 + 3: $p \longrightarrow (p \longleftrightarrow q)$, 2 + 3 $\neg p \longrightarrow \neg (p \longleftrightarrow q)$

$\equiv$

∴ We know $\boxed{p \longleftrightarrow (p \longleftrightarrow q). \text{ is } T.}$ ☆

Solve: q is either T or F. (What can q be given A?)

Case 1: q is F.

$\therefore$ $(p \leftrightarrow q) \equiv (p \leftrightarrow F)$ ← for all values of p

$\equiv \neg p$.

$\therefore$ $p \leftrightarrow (p \leftrightarrow q) \equiv p \leftrightarrow \neg p$

$\equiv F$, contradiction.

Case 2: q is T.

$\therefore$ $(p \leftrightarrow q) \equiv (p \leftrightarrow T) \equiv p$. ← for all values of p

$\therefore$ $p \leftrightarrow (p \leftrightarrow q) \equiv p \leftrightarrow p \equiv T$.

$\therefore$ Bob is a Knight. $\square$

Remark: We solved the question: given A is true, what can we say about the truth of q?

[we could also have used a truth table as in Method 1]